

# Git Workflow - Feature Error Handling UI

---

## Branch Information

- **Branch Name:** `feature/error-handling-ui`
  - **Lead:** Frontend Lead
  - **Status:** Ready for Git commands
- 

## Git Commands - Copy & Paste

### Step 1: Create & Checkout Branch

```
git checkout -b feature/error-handling-ui
```

### Step 2: Stage All Changes

```
git add .
```

### Step 3: Check Status

```
git status
```

### Step 4: View Changes (optional)

```
git diff --cached
```

### Step 5: Commit Changes

```
git commit -m "feat(frontend): implement error handling UI for 503 service
unavailable"

- Add ServiceStatusBanner component dengan color-coded styling (auth/task/all)
- Update LoginPage dengan 503 error handling dan retry button
- Update CreateTask dengan service unavailable banner
- Update EditTask dengan retry mechanism
- Update TaskList dengan service status tracking
- Add global health check di App.jsx setiap 30 detik
- Enhanced api.js dengan consistent response structure { status, data, error }
```

- Created useApi hook untuk retry logic (exponential backoff)
- Add comprehensive testing guide dan developer documentation

Fixes #feature-error-handling-ui  
Closes #task-frontend-resilience"

## Step 6: View Commit

```
git show --stat
```

## Step 7: Push Branch to Remote (when ready)

```
git push origin feature/error-handling-ui
```

## Step 8: Create Pull Request on GitHub

```
# Open: https://github.com/your-repo/pulls  
# Create PR from feature/error-handling-ui → main  
# Title: feat(frontend): implement error handling UI for 503 service  
unavailable  
# Description: See commit message
```

---

## Commit Message Template

Jika ingin customize commit message, gunakan template:

```
feat(frontend): implement error handling UI for 503 service unavailable  
  
[DESCRIPTION]  
Implementasi error handling UI untuk menangani 503 Service Unavailable  
responses  
dengan retry mechanism dan service status banner di semua pages.  
  
[CHANGES]  
- ServiceStatusBanner component untuk visual feedback  
- Enhanced api.js dengan response structure { status, data, error }  
- useApi hook dengan exponential backoff retry logic  
- Global health check di App.jsx (30 detik interval)  
- Updated LoginPage, CreateTask, EditTask, TaskList dengan error handling  
- Comprehensive testing & developer documentation  
  
[TEST COVERAGE]
```

- LoginPage: Register & Login dengan 503
- CreateTask: POST /tasks dengan 503
- EditTask: GET & PUT /tasks/{id} dengan 503
- TaskList: GET tasks, DELETE, COMPLETE dengan 503
- App.jsx: Global health check untuk auth service

#### [DOCUMENTATION]

- docs/Frontend-Error-Handling-Summary.md
- docs/frontend-error-handling-testing.md
- docs/frontend-error-handling-developer-guide.md
- /memories/repo/frontend-error-handling-implementation.md

Fixes: feature/error-handling-ui

---

## Files Changed Summary

### Created (3 files)

- frontend/src/hooks/useApi.js
- frontend/src/components/ServiceStatusBanner.jsx
- docs/frontend-error-handling-testing.md
- docs/frontend-error-handling-developer-guide.md
- docs/Frontend-Error-Handling-Summary.md

### Modified (6 files)

- frontend/src/services/api.js + 4 new methods, enhanced error handling
- frontend/src/pages/LoginPage.jsx + ServiceStatusBanner, retry logic
- frontend/src/pages/CreateTask.jsx + ServiceStatusBanner, disabled states
- frontend/src/pages/EditTask.jsx + ServiceStatusBanner, retry mechanism
- frontend/src/components/TaskList.jsx + ServiceStatusBanner, error handling
- frontend/src/App.jsx + Global health check

**Total Files Changed:** 11

**Lines Added:** ~1,500+

**Lines Deleted:** ~200

---

## Pre-Commit Checklist

Sebelum commit, pastikan:

- ☐ Code sudah tested di local environment
- ☐ No console.error atau console.warn yang tidak perlu
- ☐ No hardcoded values (gunakan environment variables)
- ☐ Component props documented dengan JSDoc comments
- ☐ Error handling pattern consistent di semua components
- ☐ ServiceStatusBanner styling responsive di mobile/tablet/desktop

- ☐ No unused imports atau dead code
- ☐ Git status clean (no untracked files yang tidak diperlukan)

Commands untuk Validation:

```
# Check for console.log di production code
grep -r "console\." frontend/src --include="*.jsx" --include="*.js" | grep -v
"node_modules"

# Check for unused variables (need eslint configured)
npm run lint

# Check build status
npm run build
```

---

## After Commit

Next Steps:

### 1. Push to Remote

```
git push origin feature/error-handling-ui
```

### 2. Create Pull Request on GitHub

- Title: `feat(frontend): implement error handling UI for 503 service unavailable`
- Description: Copy dari commit message
- Assign reviewer: Team Lead Backend / Tech Lead

### 3. Review & Feedback

- Address reviewer comments
- Update branch jika diperlukan

```
git add .
git commit --amend # jika perubahan kecil
git push --force-with-lease # push ulang
```

### 4. Merge ke Main

```
# Option 1: Merge dari GitHub UI (recommended)
# Click "Merge Pull Request" button
```

```
# Option 2: Merge dari CLI
git checkout main
git pull origin main
git merge feature/error-handling-ui
git push origin main
```

---

## Important Notes

 RULE dari AGENTS.md

### **JANGAN COMMIT DAN PUSH TANPA PERSETUJUAN USER!**

Sesuai dengan rule di AGENTS.md:

1. ☒ I already prepared the code
2. ☒ I created testing & documentation
3. ☒ I will NOT commit or push without your approval
4.  Wait for user approval before running git commands

---

## How to Review Changes

### Option 1: Review di VS Code

```
git diff HEAD~0 # Show all changes
git diff HEAD~1 frontend/src/components/ServiceStatusBanner.jsx # Specific
file
```

### Option 2: Review di GitHub

Setelah push, GitHub akan show:

- Files changed
- Diff view (green untuk added, red untuk removed)
- Comment interface untuk discussion

### Option 3: Review via Terminal

```
git log --oneline -5          # Recent commits
git show HEAD --stat         # Current commit summary
git diff --staged            # Staged changes
```

---

## Troubleshooting

### Issue: "fatal: not a git repository"

```
# Verify you're in project root
cd c:\Users\marth\cc-kelompok-a-ethereal
git status
```

### Issue: "branch already exists"

```
# If branch exists locally
git checkout feature/error-handling-ui # switch to existing branch

# Or delete and recreate
git branch -D feature/error-handling-ui
git checkout -b feature/error-handling-ui
```

### Issue: "Changes not staged for commit"

```
# Stage changes
git add .

# Or stage specific file
git add frontend/src/components/ServiceStatusBanner.jsx
```

### Issue: "Permission denied" saat push

```
# Update git credentials atau use SSH key
git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"

# Or use SSH key (recommended untuk production)
ssh-keygen -t rsa -b 4096 -f ~/.ssh/id_rsa
```

---

## Next Actions

1. **Review this guide** - Pastikan semua sesuai
  2. **Test implementation** - Run di browser
  3. **Approve for commit** - User approval needed
  4. **Run git commands** - Based on approval
  5. **Create PR** - For team review
-

**Ready for:** ⌚ User Approval

**Branch:** feature/error-handling-ui

**Status:** Implementation Complete ☑

---

## Quick Command Reference

```
# Full workflow in one copy-paste (after approval):
git checkout -b feature/error-handling-ui
git add .
git commit -m "feat(frontend): implement error handling UI for 503 service
unavailable"

- Add ServiceStatusBanner component
- Update LoginPage, CreateTask, EditTask, TaskList dengan error handling
- Global health check di App.jsx
- Enhanced api.js dengan consistent response structure
- Comprehensive testing & developer documentation"
git push origin feature/error-handling-ui
# Then create PR on GitHub
```